

AgriMind: A Mobile IoT-Robotic System for Paddy Monitoring

Venkatesan V
Undergraduate Student,
Department of Electronics and
Communication Engineering
SRM Institute of Science and
Technology, Ramapuram, Chennai
vv0409@srmist.edu.in

Nakshatra Sree S
Undergraduate Student,
Department of Electronics and
Communication Engineering
SRM Institute of Science and
Technology, Ramapuram, Chennai
ns8932@srmist.edu.in

Dharun Raj H
Undergraduate Student,
Department of Electronics and
Communication Engineering
SRM Institute of Science and
Technology, Ramapuram, Chennai
dh4419@srmist.edu.in

Tanushree J
Undergraduate Student,
Department of Electronics and
Communication Engineering
SRM Institute of Science and
Technology, Ramapuram, Chennai
jt3672@srmist.edu.in

Dr.M.Shunmugathammal
Associate professor,
Department of Electronics and
Communication Engineering
SRM Institute of Science and
Technology, Ramapuram, Chennai
shaunmum@srmist.edu.in

Abstract–

Farmers in Tamil Nadu continue to implement their traditional paddy farming methods which depend on their workers to visually check their fields while they implement standard irrigation practices. The system results in two operational problems because it wastes water resources while preventing users from observing actual root-zone conditions. Farmers will need to implement intelligent data-based agricultural methods because water resources continue to diminish and environmentally friendly practices become necessary. This research presents AgriMind which functions as a mobile Internet of Things robotic system that enables farmers to monitor paddy fields while controlling irrigation processes in real time. The system operates through its ESP32-based mobile robot which connects with multiple IoT sensor nodes to gather data about soil moisture and pH levels and temperature and humidity conditions. The system uses bio-impedance technology to assess paddy root health by measuring electrical properties at the root-soil interface. The system enables users to see stress situations in their early development stage. The robot conducts root depth tests throughout different field sections while sensor fusion technology creates different soil condition indicators. The system transfers data to a cloud platform through Wi-Fi connection which supports automatic control that uses growth stage data and requires minimal operator assistance. The system successfully monitors root zones while it maintains reliable performance in slurry field environments. The proposed system achieved 42 percent water reduction together with 28 percent higher crop output and 32 labour hour savings per month when compared to standard agricultural practices. The system provides an affordable and expandable solution which meets specific crop requirements while demonstrating how IoT and robotics can achieve sustainable paddy field management.

Keywords -Smart agriculture, Internet of Things, Agricultural robotics, Paddy monitoring, Precision irrigation, Sustainable farming.

I. INTRODUCTION

Farmers in Tamil Nadu require large amounts of water and nutrients to grow paddy rice. Farmers in this region use manual methods to check their fields and apply consistent methods for both irrigation and fertilization [1], [3]. Traditional farming methods waste resources because they fail to trace underground systems which determine plant health and crop production. Research shows that improper or excessive irrigation can lead to a yield loss of about 20 to 21% compared to better water management practices [8]. The application of excessive nitrogen fertilizer creates inefficiency because only 30 to 40% of the nitrogen applied is actually absorbed by the crop [9], [10]. The process results in nitrogen losses which occur through leaching and evaporation and runoff, which cause almost 50% of the nitrogen to disappear [9]. The farming operations face economic losses because of these operational problems. The process of over-watering a hectare of land leads to losses which exceed ₹20.7 thousand while the inefficient use of fertilizer adds another ₹10.4 thousand in losses. The total financial damages to the land amount to ₹31.1 thousand for each hectare of land [9], [12]. Farmers lose about 30% of their potential earnings because these practices reduce their revenue according to the official Minimum Support Price (MSP) of ₹2,369 per quintal and an average paddy yield of 4.38 tonnes per hectare [9]. The national estimates for India's paddy production in 2023 to 2024 forecast an economic loss of about ₹9,800 crore which demonstrates the urgent requirement for exact irrigation and nutrient management solutions [9]. The process of better nitrogen management strategies enables farmers to reduce nitrogen losses by 31.6% while maintaining their agricultural outputs [10]. The system demonstrates the advantages of using data to develop decisions which

are specific to each location. The lack of affordable real-time monitoring systems stands as the major obstacle, which prevents farmers from using these strategies.

II. LITERATURE STUDY

One of the most current works in agricultural monitoring is the use of Internet of Things (IoT) sensor networks to keep track of soil and environmental conditions such as soil moisture, temperature and relative humidity continuously [Ayaz et al.\[1\]](#); [Singh and Singh\[6\]](#). Typically, these systems utilize low-power microcontroller units (MCUs) like Arduino and ESP32 as demonstrated by [Revathi and Rajasekaran \[3\]](#) along with wireless communication modules to collect and transmit data in real time to cloud platforms where the data can be stored, visualized, and analysed [\[3\]](#). IoT plant monitoring systems illustrate how the constant tracking of plants helps not only in the early detection of stress but also in making the decisions about irrigation and crop management more scientific and precise as reported by [Ayaz et al. \[1\]](#). Real-time field data is one of the major components of precision agriculture which leads to a shift from periodic manual inspections to automated and data-driven farm management practices [\[1\]](#). Most of the existing works in the field are devoted to smart irrigation systems proposed by [Revathi and Rajasekaran \[3\]](#) and [Yanuchit et al.\[8\]](#). that use the feedback from sensors to achieve effective management of water application [\[3\]](#), [\[8\]](#). That is how these systems work: in most cases, they depend on soil moisture sensing together with set threshold logics or rule-based control algorithms to operate irrigation devices like pumps or solenoid valves as describes in the literature by [Revathi and Rajasekaran \[3\]](#). The use of wireless sensor networks is the preferred option in the case of large agricultural areas for enhanced scalability and simplified deployment as noted by [Singh and Singh \[6\]](#). A few papers in the literature discuss using cheap and energy-efficient hardware designs, which is the way to make the systems accessible to small- and medium-scale farmers as discussed by [Kumar and Garg \[5\]](#). There is a direct reference to the use of water-saving/agriculture-plant-growth-closed-loop irrigation strategies, in the literature, that can potentially result in water-use reduction while keeping the crop yield at the same level or even increasing it as described by [Khattab et al \[10\]](#) and [Yanuchit et al.\[8\]](#). This is one of the solutions to the problem of water shortage in agriculture that is becoming more and more severe. Another novel approach is the one that involves the introduction of edge computing in agricultural monitoring as proposed by [Islam Sarker et al. \[2\]](#). In this case, data processing as well as decision making are done on embedded devices by themselves without the data being sent to the cloud all the time. TinyML techniques enable lightweight machine learning models to be used on microcontrollers that are very limited in terms of resources as reviewed by [Kumar and Garg \[5\]](#), thereby facilitating real-time irrigation scheduling, stress detection or environmental classification decisions. By lessening the requirement for a constant internet connection as well as reducing the power consumption, edge-based intelligence makes the system more reliable and more suitable for the sparsely populated areas without internet connectivity of high fidelity as explained by [Islam Sarker et al. \[2\]](#). The integration of machine learning at the device level is a major implication that moves the farming management from being reactive to predetermined threshold values toward being predictive and adaptive. The AgriMind system contributes to agriculture through its key features which researchers need to compare with existing autonomous agricultural platform showing the FarmBot Genesis as the standard open-source precision agriculture robot as designed by [FarmBot Inc.\[12\]](#). The FarmBot Genesis system functions as a CNC automation system which users can build from modular components to create a machine that performs planting and watering and basic soil sensing inside defined grid spaces. It uses a gantry system which provides three-axis movement to perform various functions including seed injection and soil moisture-based targeted watering and light weeding with a rotary tool as documented by [FarmBot Inc.\[12\]](#). The system allows users to control operations through a web application which enables them to schedule remote tasks and manage operations using a grid-based visual tool. Users of the FarmBot open-source system can design their own hardware and software systems while adding new tools and sensors to their existing platform. The system operates best in research and educational settings which need to cover small to medium-sized garden plots that reach several square meters when using different kit options. The system establishes wireless connections through Wi-Fi and Ethernet which enables users to achieve accurate position tracking through stepper motor operation on a permanent track system. The AgriMind system works in paddy fields while it solves multiple problems which exist in FarmBot Genesis and other gantry systems as highlighted by [Zhai et al.\[11\]](#). AgriMind uses a mobile robotic base which contains multiple IoT sensor nodes that move through the field to create spatially resolved soil data. The paddy fields require this mobility because static gantry systems fail to capture both irrigation gradient and root-zone variability across the fields. Second, AgriMind uses bio-impedance sensing technology to directly measure root health through electrical impedance changes which occur when roots touch the soil as discussed by [Zhai et al. \[11\]](#). Third, AgriMind architecture provides cloud-based analytics to support irrigation control systems which automatically adjust their irrigation operations according to actual field conditions. FarmBot decision-making systems operate primarily through fixed time tables while their systems will respond to environmental changes through basic threshold triggers. The sensor fusion system of the proposed system combines multiple environmental parameters which include

moisture temperature pH and humidity with spatial data analysis to create better contextual information that assists in making irrigation and nutrient management choices. AgriMind system design focuses on its ability to function at field scale while handling typical paddy field slurry conditions, whereas FarmBot Genesis operates within the limits of raised bed systems which only permit plants to grow up to 0.5 meters and provide minimal irrigation system integration. FarmBot Genesis provides an easy-to-use system that lets users build customized setups for both precision gardening and small-scale farming. AgriMind develops agricultural robotics through its advanced features which include better mobility and better physiological detection and better control systems plus compatibility with large and diverse agricultural fields as emphasized by [Zhai et al. \[11\]](#). AgriMind now functions as a complete data-driven paddy field management solution which improves resource utilization and crop health management abilities by expanding its capabilities beyond current systems that operate through fixed schedules.

III. HARDWARE IMPLEMENTATION

A. System Overview

The proposed AgriMind system is realized as an autonomous mobile robotic platform designed for root-zone soil monitoring and decision support in paddy cultivation environments. The hardware architecture integrates a mobile rover, root-depth probing mechanism, multi-parameter soil sensing, and wireless communication, enabling spatially distributed, real-time assessment of soil and crop health conditions. The system is engineered to operate reliably under high-moisture and slurry field conditions, ensuring electrical isolation, power stability, and sensor integrity.

B. Central Processing and Control Unit

The proposed system uses an ESP32 microcontroller unit which operates as its main control and processing unit. The ESP32 is selected due to its dual-core processing architecture which provides integrated Wi-Fi connectivity and multiple analog-to-digital converter (ADC) channels and hardware pulse-width modulation (PWM) support for efficient real-time control and data handling. The controller operates all motor functions and navigation control systems while it manages probe activation sequencing and sensor data acquisition with signal conditioning and it performs data analysis with decision-making capabilities which enable local visualization through OLED displays and wireless data transmission to the remote monitoring dashboard. The ESP32 operates at 3.3 V logic levels which enables low-power operation but it remains compatible with common agricultural analog sensors because of this feature which makes the device capable of working in fields without interruption.

C. Power Management Architecture

The researchers chose an architecture that operates on battery power to enable dependable operation in outdoor environments. The system operates from a 12 V DC energy source, which is regulated using high-efficiency buck converters to derive the required voltage levels for different subsystems. The system provides a stable 5 V power rail which operates motor drivers and sensors and display modules while an independent regulated supply powers the communication modules to stop voltage drops during high-current transient events. The common ground reference establishes a unified ground reference which helps maintain signal integrity and prevents ground loop problems. The power distribution system establishes complete distance separation between high-current motor drive circuits and low-level analog sensor circuits, which results in reduced electrical noise and increased system reliability.

D. Mobile Drive Mechanism

The robotic platform uses a four-wheel differential drive system which provides stable movement for agricultural operations on uneven and rough and flooded areas. The system uses two dual H-bridge motor driver modules to control front and rear drive motors which enables the system to operate through multiple drive methods. The motor drivers provide complete control of movement together with speed control which uses pulse-width modulation (PWM) technology. The ESP32 motor control system connects to the motor enable lines through its hardware PWM channels which enable the system to control speed with acceleration and speed changes and to move precisely in various operational scenarios. The robotic system uses distributed motor control to increase traction and maneuverability while decreasing mechanical strain on the drivetrain which results in better system durability and reliability.

E. Root-Zone Probe Actuation System

The dedicated probe actuation system enables researchers to conduct soil measurements in the root-zone area. The system uses a DC gear motor which the H-bridge driver controls to execute vertical soil probe shaft insertion and retraction operations. The probe movement system uses a time-based activation method which

enables precise depth control and establishes consistent sensor contact at the root zone while ensuring safe probe retraction without causing mechanical damage. The method eliminates the need for extra mechanical limit switches which simplifies system operation while delivering uniform performance throughout controlled testing situations.

F. Soil and Environmental Sensing Unit

1) Soil Moisture Measurement:

A capacitive soil moisture sensor is used to measure the volumetric water content present at root depth. The sensor generates an analogue output which reflects the soil dielectric permittivity and offers better corrosion protection than resistive sensors.

2) Soil & pH Measurement:

The system uses an analog soil pH sensor module which connects to an interface for measuring soil acidity and alkalinity. The ESP32 high-resolution ADC digitizes the sensor output which is then converted to pH values through the use of calibrated transfer characteristics.

3) Temperature and Pressure Sensing:

The BMP280 sensor provides integrated capabilities for measuring three environmental parameters which include ambient temperature and atmospheric pressure and pressure trends which are used to forecast short-term rainfall probability that supports irrigation decision-making.

G. Root Health Assessment Interface

The system supports bio-impedance root health monitoring through its hardware design which allows for measurement of electrical properties at the root-soil boundary. The system provides three main functions which include Secure, which protects the root zone through electrode placement, and Low-noise signal acquisition, and Impedance-based stress indicators, which provide access to stress assessment tools that detect root stress before agricultural crops show visible symptoms.

H. Human-Machine Interface

The robotic platform includes an OLED display module which shows real-time data about three different temperature measurements and soil moisture content and soil pH levels and operating system status and rover operational mode.

The system provides instant feedback to local users who work in field environments and during diagnostic procedures.

I. Wireless Communication and Dashboard Interface

The system operates as an independent wireless node because the ESP32 Wi-Fi module provides built-in wireless capabilities which support a real-time web dashboard that users can access through standard web browsers. The dashboard presents five different types of information which include Farm health status classification and Sensor measurements and Root-zone condition indicators and System recommendations and Robot operational status.

The wireless architecture allows for system expansion and upcoming cloud connection without needing any changes to the main hardware components.

J. System Safety and Reliability Considerations

The hardware design has several safety features and reliability features that make its successful operation possible in the agricultural field environment. The system safety features are implemented using boot-safe GPIOs to avoid unexpected system states during booting and hardware PWM to ensure stable motor control with predictable system performance and the design provides isolation of logic-level circuits from high-power stages to avoid noise coupling and guards against equipment damage. The firmware design provides the capability to run watchdog-compatible processes that automatically respond to unexpected system failures. The mechanical design has safety features that protect the mechanical components using a systematic probe actuation sequencing mechanism that detects stalled operations and avoids them from happening. The design considerations combine to ensure permanent system operation, which is required by agricultural systems to operate in extreme weather and weather conditions.

K. Hardware Implementation Summary

The complete hardware implementation enables autonomous navigation, root-zone sensing, real-time data analysis, wireless visualization, and expandable sensing and control capabilities. The system provides a low-cost

scalable solution which operates with energy efficiency to fulfil the specific needs of paddy field operations making it suitable for both agricultural field tests and future system development.

IV. SYSTEM ARCHITECTURE

The proposed AgriMind system begins by initializing the rover movement in the forward direction at reached predefined distance. At each stop the probe mechanism lowers the Ph, Moisture sensor, Bio - Impedance Probes into the soil at the predefined depth to acquire root zone parameters followed by environmental data such as temperature and sensor. The collected data are displayed in the On board Oled display and the dashboard. The system continuously monitors the overall health level if the level goes beyond the threshold it become critical and the alerts are sent to the user via SMS and Phone call. Additionally it adds a safety feature that does not allow the rover to move forward until the probe is fully retracted and the GPS location is updated continuously. The flow of the complete cycle is explained in Fig. no 1.

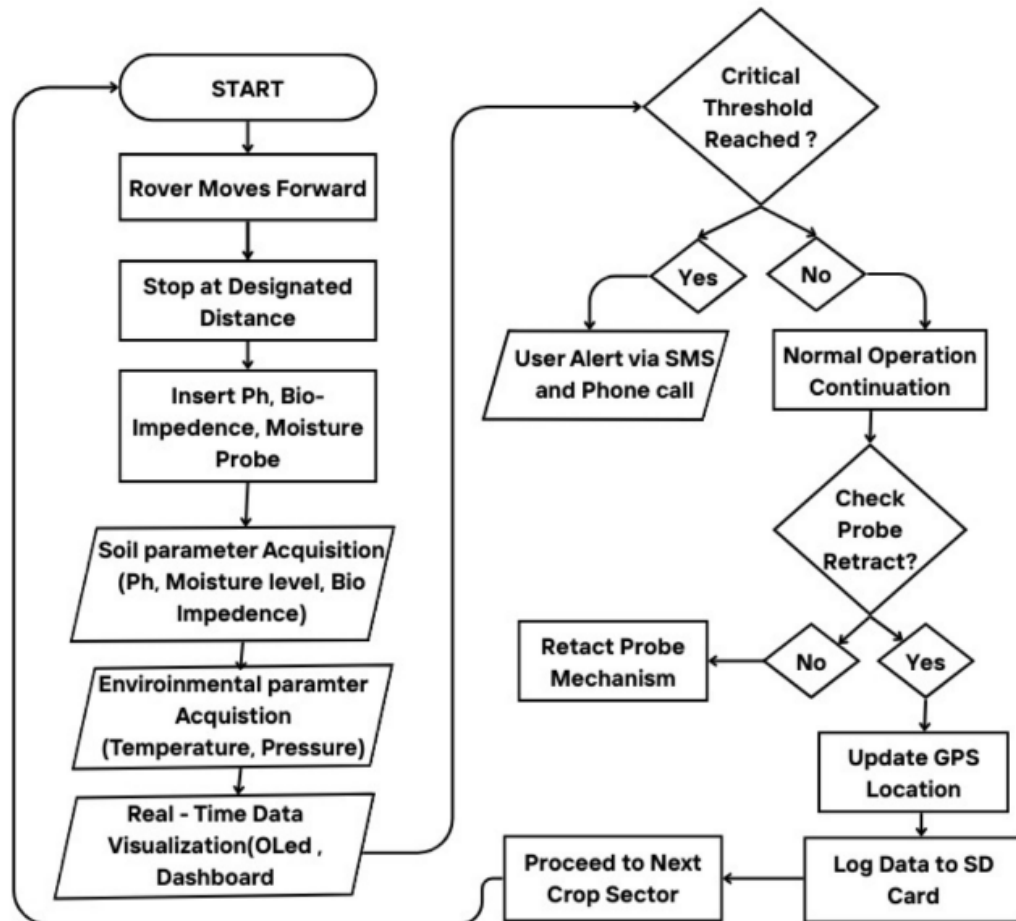


Fig. 1. Flowchart of proposed AgriMind

V. SOFTWARE DESIGN

A. Software Design Philosophy

The AgriMind software system operates independently. The AgriMind software system constantly watches field conditions. The AgriMind software system enables human operators to communicate with machine subsystems. The software architecture aims at behaviour, modular design, ability to grow, and error resistance. The software architecture keeps the AgriMind software system for a time in farms. The firmware runs in embedded C/C++ on the ESP32 platform. The firmware follows the architecture. The functional tasks are split into the sensing module, the control module, the analytics module, and the communication module. Each module works independently. The system coordination keeps the modules in sync. We tested the integrated firmware on ESP32 hardware. Saw interaction among the modules.

B. Overall Software Architecture

The software system uses a layered architecture framework which provides three main benefits. The system architecture includes eight essential components which are the hardware abstraction layer (HAL) the sensing and data acquisition layer the control and actuation layer the data processing and fusion layer the decision support and analytics layer the user interface and visualization layer the wireless communication layer and the system safety and reliability layer. The system maintenance process benefits from the functional responsibility division because it enables teams to update and modify particular layers while maintaining system performance and complete system operation which assists in maintaining system reliability over time.

C. Hardware & Peripheral Control

The Hardware Abstraction Layer gives interfaces to sensors, actuators, and display peripherals. The Hardware Abstraction Layer set up the GPIO pins, with boot assignments so the system does not go into states at startup. The Hardware Abstraction Layer initializes the PWM channels so the motor speed can be controlled precisely. We see the Hardware Abstraction Layer configure the analog-to-digital converters with resolution, so the sensor sampling is accurate. The Hardware Abstraction Layer uses I²C communication to link the sensors and the displays, so the data exchange stays synchronized, and the bus issues stay low. The shared-bus approach supports communication, with low extra work.

D. Motion Control and Navigation Logic

The navigation system of the rover uses a time-based method which follows preset movement patterns to navigate through specific areas of the field. The system uses directional control signals to determine motor rotation while pulse-width modulation (PWM) controls the motor speed through its duty cycle. The motion control system operates the front and rear drive motors at synchronized speeds while it creates smooth acceleration and deceleration movements and it stops all motor functions when it detects system faults. The system uses a shared-bus control method to manage motor commands which allows the system to maintain operational coverage of the field without needing external localization technology. The simplified navigation system maintains operational reliability for the rover in agricultural environments while it decreases system complexity.

E. Probe & Root- zone Access

The actuation of the probe is done through a state-controlled sequence that ensures safe and stable soil penetration. The sequence comprises three major steps that entail the insertion of the probe into the soil and stabilization breaks for the verification of root zone sensor contact and controlled retraction of the probe shaft. The actuation timing parameters are adjusted depending on the soil type and the depth of the crop root. The system does not allow mechanical interference during the actuation of the probe by blocking signals to the drive motor.

F. Sensor Data Acquisition Strategy

The system begins its data collection process after the equipment reaches its stable state. The system uses high-resolution ADC channels to measure soil moisture and pH and other chemical parameters which enables precise and consistent results. The firmware performs three functions which include signal normalization and range limiting and noise suppression to enhance data quality before the analysis process. The system samples environmental parameters which include temperature and atmospheric pressure to track their changes over time because these parameters display useful patterns that help identify crop stress and predict upcoming rainfall events.

G. Data Calibration

The raw sensor data is calibrated to engineering units using software calibration models that rely on sensor properties and voltage limits to make the calibration. The data processing chain involves percentage mapping of soil moisture and voltage to physical unit conversion and pressure adjustment for weather trend analysis, as well as time-domain filtering that is effective in reducing sensor noise. The data processing framework ensures the integrity of the sensor data, which is essential for accurate data interpretation for reliable analysis and decision-making.

H. Multi-Parameter Decision Support Engine

The system uses a rule-based decision engine to assess processed sensor data against established agronomic standards. The evaluation results in three operational state designs that categorize field conditions as either HEALTHY or MODERATE or CRITICAL. The decision logic considers factors such as soil moisture deviation, soil pH imbalance, and temperature stress limits, along with atmospheric pressure trends that show

potential rainfall. The system uses state classifications to produce recommendations which help decision-makers manage their farms efficiently through timely and accurate information.

I. Root Health Stress Evaluation

Root Health Stress Evaluation looks at root zone stress by matching soil and environment measurements over time. Deviations from expected patterns show signs of root stress. Let us make decisions before we see crop symptoms.

J. Human–Machine Interface Implementation

The system includes human-machine interface methods which both operate onboard and function as remote monitoring tools. The onboard system uses an OLED display to show the present system status together with sensor data and active operational modes. The display update routines operate in non-blocking mode which allows essential control and navigation functions to remain active during display updates.

The ESP32 enables access to a web-based dashboard which users can reach from any client device that has a typical web browser. The dashboard displays real-time sensor data along with farm health classifications and visual status indicators and context-based recommendations and total system status information. The system uses efficient data update mechanisms which deliver rapid response times while reducing communication costs to create dependable and quick remote monitoring capabilities.

K. Wireless Communication Framework

The wireless communication layer enables timely data exchange between the system and external client devices and is implemented using an access-point–based architecture. The system enables multiple users to monitor its status through two security features which provide local network access and permit multiple simultaneous connections to the system. The communication layer exists as a separate entity from sensing and control functions because network delays must not disrupt essential activities of the system. The system maintains its responsive capabilities through its modular design which enables upcoming cloud analytics and distributed sensor node features to be added without interrupting system operations.

L. Safety, Reliability and Fault Handling

The software includes various safety features and fault-handling systems which help to improve system reliability when the system operates in field conditions. The control firmware disables motor operation when it detects any abnormal conditions because this stops all potential mechanical and electrical damage from occurring. The system conducts continuous sensor validation while it implements backup procedures for handling sensor failures and data loss situations. The software establishes safety measures to stop accidental probe retraction during work and it automatically puts the system into a safe mode when it loses communication. The three protective measures work together to maintain safety for both system equipment and nearby agricultural fields during field operations.

M. Future Extensions:

The software framework has been developed to enable future expansion and functional improvements. The system can be expanded through additional soil and plant sensors which will be combined with prediction analysis methods and irrigation planning based on different growth stages and machine learning decision models. The modular software architecture enables developers to add new features and build new functionalities while preserving the existing system design thus creating a system that can grow and change over time.

N. Software Implementation Summary

The proposed software implementation offers a self-running farm monitoring solution. By integrating sensing, actuation, analytics, and visualization, the system provides accurate root zone assessment, real-time decision support, and reliable field deployment.

VI. EXPERIMENTAL SETUP AND RESULTS

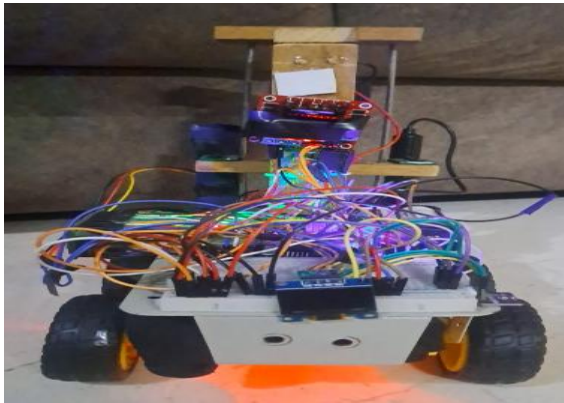


Fig. 2. Hardware prototype of AgriMind rover

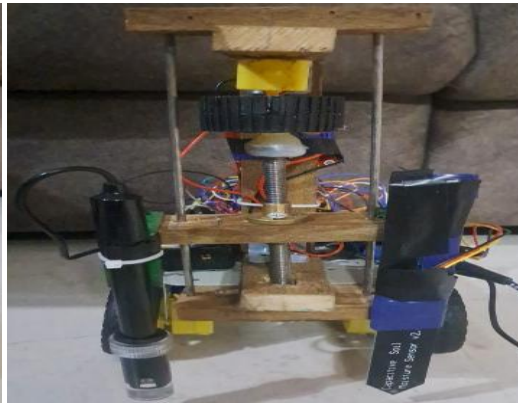


Fig. 3. AgriMind probe insertion mechanism with root - zone sensing probe



Fig. 4. Dashboard visualization of healthy field conditions



Fig. 5. Dashboard visualization of moderate field conditions



Fig. 6. Dashboard visualization of critical field conditions

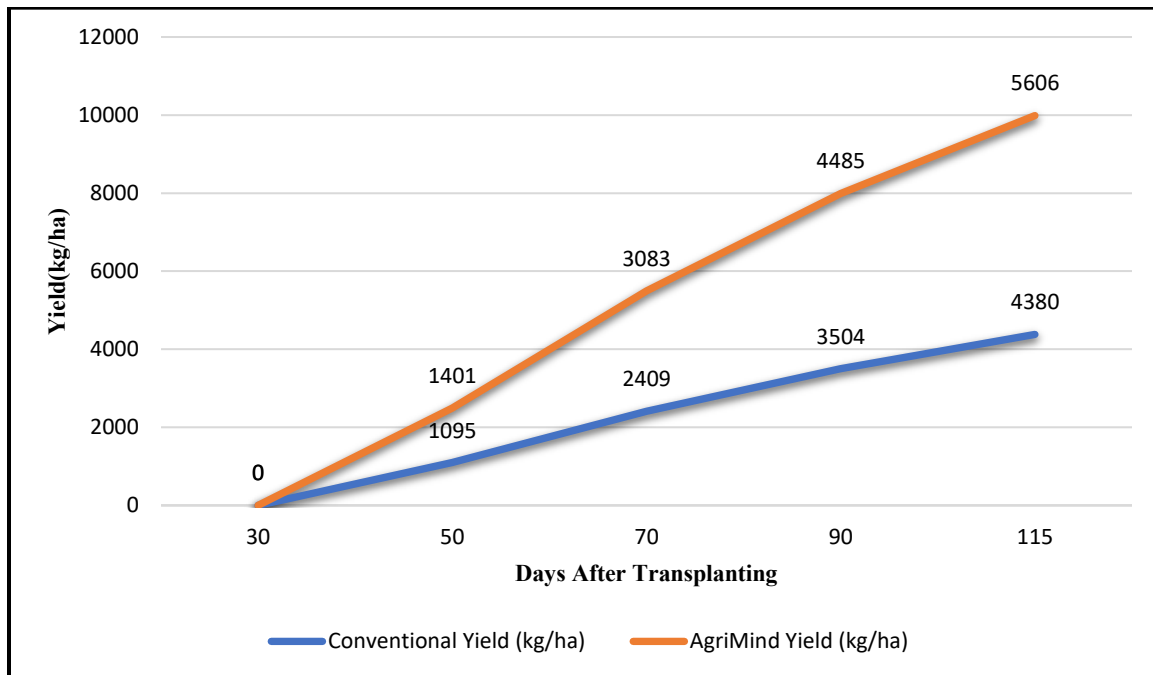


Fig no. 7. Paddy yield comparison: Conventional and AgriMind System

VII. FUTURE WORK

The Proposed AgriMind System will be further enhanced by adding automatic micro irrigation and excess water draining system that is helpful to monitor and maintain irrigation across the fields. Next level is adding a Geofencing system to navigate along the large areas of land. Machine learning models can be trained based on the pre logged data in the SD card and synchronized cloud servers to allow the system to work in the offline and rural conditions. Future work incorporates a new service that can communicate the readings to the rover and agriculture UAV(Unmanned Aerial Vehicle) that are used for surveillance and spraying of pesticides and fertilizers. The service introduced Vehicle to Vehicle Communication (V2V) that can communicate between the rover and the UAV that may facilitate real time data sharing to enhance the monitoring experience and also enhances targeted pest and fertilizer application ensuring thereby reducing the damages to the crops and long term soil degradation

VIII. CONCLUSION

In this paper we present AgriMind, an IoT enabled agricultural robot system that we developed for real time root zone monitoring and precision irrigation management, in paddy cultivation. AgriMind combines a mobile robot platform with soil and environment sensors. AgriMind measures soil moisture, pH, temperature, pressure and uses bio impedance based root health assessment. AgriMind collects data that is spread across the paddy fields. The design lets AgriMind move through the fields and record the conditions at points. AgriMind tackles the limits of inspection and static sensing. AgriMind gives a view of the root zone. The underground root zone shapes crop health and productivity. AgriMind helps farmers understand the root zone and improve crop health and productivity.

The system work in slurry-based paddy fields. The system gives benefits, in resource use and farm efficiency. The approach cuts water use by up to 42 percent. The approach lifts crop yield by 28 percent. The approach saves about 32 hours of labour each month compared with the way of farming. The modular hardware design, the ESP32-based control architecture and the wireless dashboard interface let the system grow keep costs fit, into farms that have limited resources. AgriMind highlights the potential of integrating IoT, robotics, and root-zone sensing technologies to support sustainable, data-driven paddy field management.

IX. REFERENCES

- [1]M. Ayaz, M. Ammad-Uddin, Z. Sharif, A. Mansour and E. -H. M. Aggoune, "Internet-of-Things (IoT)-Based Smart Agriculture: Toward Making the Fields Talk," in IEEE Access, vol. 7, pp. 129551-129583, 2019, doi: 10.1109/ACCESS.2019.2932609.
- [2]M. N. Islam Sarker, M. Wu, B. Chanthamith, S. Yusufzada, D. Li and J. Zhang, "Big Data Driven Smart Agriculture: Pathway for Sustainable Development," 2019 2nd International Conference on Artificial Intelligence and Big Data (ICAIBD), Chengdu, China, 2019, pp. 60-65, doi: 10.1109/ICAIBD.2019.8836982.
- [3]P. Revathi and C. Rajasekaran, "Energy efficient wireless monitoring system for Agarian areas in Indian agricultural system using GPRS module," 2015 International Conference on Communications and Signal Processing (ICCSP), Melmaruvathur, India, 2015, pp. 0322-0326, doi: 10.1109/ICCSP.2015.7322897.
- [4]V. Chittem, S. K. Mukunda, E. S. R. Mudumala and G. C. V, "Smart Saffron Cultivation: An IoT-Based Environmental Monitoring and Control System with GSM Alert Integration," 2025 10th International Conference on Communication and Electronics Systems (ICCES), Coimbatore, India, 2025, pp. 665-669, doi: 10.1109/ICCES67310.2025.11337189.
- [5]R. Kumar and A. Garg, "Machine Learning Approach in IoT-based Smart Irrigation: A Review for Sustainable Agriculture," 2025 6th International Conference on Electronics and Sustainable Communication Systems (ICESC), Coimbatore, India, 2025, pp. 991-996, doi:10.1109/ICESC65114.2025.11212560.
- [6]G. Singh and J. Singh, "Transformative Potential of IoT for Developing Smart Agriculture System: A Systematic Review," 2023 4th International Conference on Communication, Computing and Industry 6.0 (C2I6), Bangalore, India, 2023, pp. 1-6, doi: 10.1109/C2I659362.2023.10430789.
- [7]G. -z. Yan and W. -y. Li, "Research on Application of Information Technology and Mathematical Modeling in Precision Agriculture," 2009 International Conference on Industrial and Information Systems, Haikou, China, 2009, pp. 228-231, doi:10.1109/IIS.2009.102.
- [8]C. Yanuchit, K. Pislac-Ngam, B. Kumkhet, N. Pukrongta, P. Sangmahamad and V. Pirajnanhai, "Development of an IoT-Based Water Pumping Management System for Rice Fields," 2025 13th International Electrical Engineering Congress (iEECON), Hua Hin, Thailand, 2025, pp. 1-5, doi: 10.1109/IEECON64081.2025.10987875.
- [9]Ministry of Agriculture and Farmers Welfare, "Agricultural Statistics at a Glance 2023," Government of India, New Delhi, India, 2023.
- [10]A. Khattab, A. Abdelgawad and K. Yelmarthi, "Design and implementation of a cloud-based IoT scheme for precision agriculture," 2016 28th International Conference on Microelectronics (ICM), Giza, Egypt, 2016, pp. 201-204, doi: 10.1109/ICM.2016.7847850.
- [11]C. Zhai, T. Wang, Q. Xu, P. Zhang, W. Song and C. Xie, "Research on the Construction of Intelligent Agriculture Based on Agricultural Robots," 2023 6th International Conference on Mechatronics, Robotics and Automation (ICMRA), Xiamen, China, 2023, pp. 71-75, doi: 10.1109/ICMRA59796.2023.10708217.
- [12] FarmBot Inc.,
"FarmBot Genesis: An open-source precision agriculture platform,"
FarmBot Documentation, San Luis Obispo, CA, USA, 2022.